- Ass 3 due Fri 9 Oct @ 4pm

Recap:

(1) We should identify parameter(s) of interest, specify null & alternative hypotheses, choose a test statistic $T(\underline{X})$, derive our decision rule $\{T(\underline{X}) \in C\}$ based on Type I error & power considerations ALL BEFORE collecting our data.

Sir R.A. Fisher: "To consult a statistician after an experiment has finished is to ask him to conduct a post-mortem examination: He can perhaps say what the experiment died of."

(2) After we collect our data, we can then make our decision based on the observed outcome of our test statistic $t_{obs} = T(\underline{x})$

(3) We can then supplement our decision by computing a p-value.

Example (Tutorial Wk 10, continued)

Recall that our 5%-level decision rule was to reject $H_0: \lambda = 7.3$ in favour of $H_1: \lambda > 7.3$ if

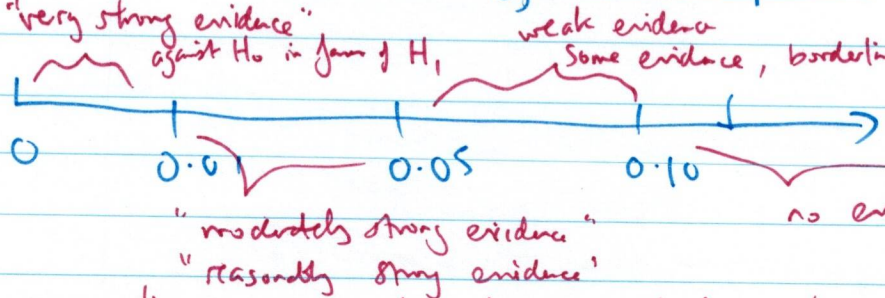
$$\{X_1 + X_2 + \cdots + X_{15} \geq 128 \text{ calls}\}$$

After collecting our data, we observed $x_1 + x_2 + \cdots + x_{15} = 132$ calls

Decision? Reject $H_0: \lambda = 7.3$ in favour of $H_1: \lambda > 7.3$

P-value to support decision: $P_{H_0}(\text{Pois}(15 \times 7.3) \geq 132) \approx 2\%$.

General thresholds & terminology used for p-values



NOTE: there are no firm boundaries between these descriptors (e.g. a p-value of 0.049 or 0.051 are both qualitatively the same)

Because p-values are often misinterpreted (("the chance that H_0 is true"), this is one of the reasons why the Journal of Applied Psychology actually banned p-values & hypothesis tests in 2014, in favour confidence intervals. Does this really improve anything?

Firstly, CIs are equally misinterpreted:
"the chance that the value is between xx & xxx is 95%." X
Secondly, there is a duality between CIs & hypothesis testing / p-values:

Duality between CIs & hypothesis testing

Given a dataset, a CI provides us with a range / set / collection of parameter values under which the data would have been likely to occur ("parameter values that are consistent with the data")

On the other hand, given a parameter value, a hypothesis test specifies a range of data values that would have unlikely to be observed

We can see that CIs are simply the inverse of hypothesis testing (& vice versa). More precisely:

- two-sided CI is precisely a set of parameter values that would be accepted as the null hypothesis in a two-sided test
- one-sided " " " " " in a one-sided test.

Example (CIs for μ when σ^2 is unknown, continued)

Recall that for testing $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$, the rejection region for a t-test is given by

$$\left\{ \bar{X} \leq \mu_0 - t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \quad \text{OR} \quad \bar{X} \geq \mu_0 + t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \right\}$$

The acceptance region is therefore:

$$\left\{ \mu_0 - t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \leq \bar{X} \leq \mu_0 + t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \right\}$$

$$\Leftrightarrow \left\{ \bar{X} - t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \leq \mu_0 \leq \bar{X} + t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \right\}$$

which is exactly the CI for μ via the t-dsgn:

$$\left(\bar{X} \pm t_{(n-1)}^{(1-\alpha/2)} S/\sqrt{n} \right)$$

When exact pivot can't be constructed, we can always construct a CI by inverting hypothesis tests (Tut week 10)

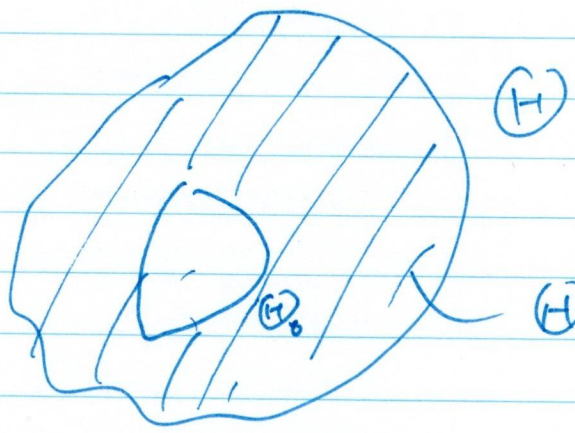
Generalized likelihood ratio test (GLRT)

The GLRT can be used to unify all the methods from §7 & §8 of your reference notes.

Likelihood ratio tests for when both the null & alternative hypotheses are NOT single point parameters. This is particularly useful when a parameter is multidimensional.

Suppose that the null hypothesis $H_0: \theta \in \Theta_0$, where Θ_0 is some subset of the full parameter space Θ.
The alternative is $H_1: \theta \in \Theta_1$, where Θ_1 is another subset of the full parameter space
(& Θ_0 & Θ_1 are disjoint)

(So if we are considering "two-sided" alternatives, we are typically thinking of "outsided" or "insided" alternative)



Consider the GLRT statistic:

$$\frac{\sup_{\theta \in \Theta_0} L(\theta \mid \underline{X})}{\sup_{\theta \in \Theta_1} L(\theta \mid \underline{X})}$$

← best possible likelihood under null model

← best possible likelihood under alt. model.

If this ratio is "small", then the data would have been more likely observed under alternative Θ_1, than under the null Θ_0. This would constitute evidence against the null Θ_0 in favor of the alternative Θ_1.

Example 7.3 (comparing two group means)

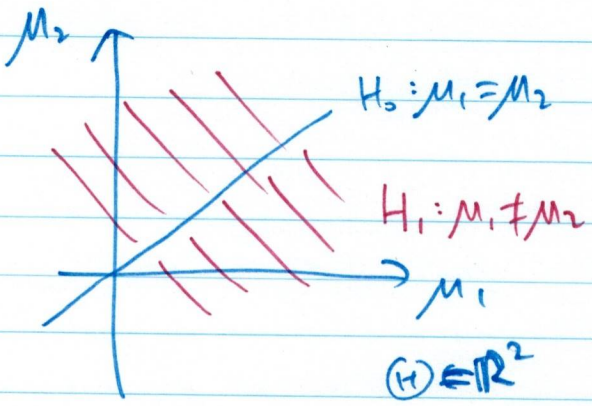
Let $\underline{X}_1 = (X_{11}, X_{12}, \ldots, X_{1n_1}) \overset{iid}{\sim} N^*(\mu_1, \sigma^2)$

& $\underline{X}_2 = (X_{21}, X_{22}, \ldots, X_{2n_2}) \overset{iid}{\sim} N^*(\mu_2, \sigma^2)$

be two indep samples, where μ_1 & μ_2 are unknown parameters, but assume σ^2 is known (for now)

We may want to test whether the two group means are the same:

$$H_0: \mu_1 = \mu_2 \quad \text{vs.} \quad H_1: \mu_1 \neq \mu_2$$



Note: neither hypotheses are single points. Rather, the null hypothesis is a one-dimensional line in a two-dimensional parameter space.

To use the GLRT, we first need the likelihood fn:

$$L(\mu_1, \mu_2 \mid \underline{X}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n_1+n_2} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n_1}(X_{1i}-\mu_1)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n_2}(X_{2i}-\mu_2)^2\right\}$$

For the numerator of the GLRT, we want to maximise $L(\mu_1, \mu_2)$ under $H_0: \mu_1 = \mu_2$. Using intuition, the MLE of the common mean is

$$\hat{\mu}_1 = \hat{\mu}_2 = \frac{\sum_{i=1}^{n_1} X_{1i} + \sum_{i=1}^{n_2} X_{2i}}{n_1 + n_2} = \bar{X}_{..}$$

grand mean

↑ "dot notation" denotes averaging over both indices.

So the maximal likelihood achieved under H_0 is:

$$\sup_{\mu_1=\mu_2} L(\mu_1, \mu_2 \mid \underline{X}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n_1+n_2} \exp\left\{-\frac{1}{2\sigma^2}\sum(X_{1i}-\bar{X}_{..})^2 - \frac{1}{2\sigma^2}\sum(X_{2i}-\bar{X}_{..})^2\right\}$$

Let's look at the expression inside the exponent:

$$\underbrace{\sum(X_{1i}-\bar{X}_{..})^2 + \sum(X_{2i}-\bar{X}_{..})^2}_{\text{Total Sum of Squares } (SS_{Total})} = \sum_{i}^{n_1}\underbrace{(X_{1i}-\bar{X}_{1.}}_{a} + \underbrace{\bar{X}_{1.}-\bar{X}_{..})}_{b}{}^2 \quad ①$$

$$+ \sum_{i}^{n_2}\underbrace{(X_{2i}-\bar{X}_{2.}}_{c} + \underbrace{\bar{X}_{2.}-\bar{X}_{..})}_{d}{}^2 \quad ②$$

$$① = \underbrace{\sum(X_{1i}-\bar{X}_{1.})^2}_{a^2} + \underbrace{2(\bar{X}_{1.}-\bar{X}_{..})\sum(X_{1i}-\bar{X}_{1.})}_{2ba=0} + \underbrace{n_1(\bar{X}_{1.}-\bar{X}_{..})^2}_{b^2}$$

$$= \sum(X_{1i}-\bar{X}_{1.})^2 + n_1(\bar{X}_{1.}-\bar{X}_{..})^2$$

② = H/W.